Functors out of a product precategory are the functors into the functor precategory

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 899-905 and its proof environment opens on line 906, the next line; the pairing is by adjacency and the retained proof range 906-923 is the whole environment.

Editorial note. The retained counter values (chapter 9, section 5, theorem offset 2) make the counted lemma print as Lemma 9.5.3, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel ct:functorexpadj = 9.5.3).

Editorial note. The author defers three routine verifications, quoted here verbatim as the project requires: "Functoriality in a is easy to check", "Functoriality is again easy to check" and "Finally, it is also clear that these operations are inverses." The main-line construction (both directions and the naturality observation) is carried out, so the card is counted under the project criterion that a complete main argument with only routine verifications left open is acceptable.

Editorial note. No context span and no external reference are needed: the retained statement and proof use no label or citation from outside their own ranges.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the exponential adjunction identifying functors out of a product precategory with functors into a functor precategory (author label ct:functorexpadj, book Lemma 9.5.3).

Allowed standard prerequisites (editorial).

- Product precategory. The product of two precategories has pairs of objects as objects and pairs of morphisms as morphisms, with the componentwise structure.. It is the domain of the functors on the left hand side.
- Functor precategory. For precategories B and C the functor precategory has functors as objects and natural transformations as morphisms.. It is the codomain on the right hand side.
- Natural transformation and functoriality. A natural transformation is a family of morphisms commuting with the action on morphisms, and a

functor must respect composition and identities.. They are what the two constructions have to produce.

Lemma 9.5.3. *For precategories A, B, C, the following types are equivalent.*

(i) *Functors $A \times B \to C$.*
(ii) *Functors $A \to C^B$.*

Proof. Given $F : A \times B \to C$, for any $a : A$ we obviously have a functor $F_a : B \to C$. This gives a function $A_0 \to (C^B)_0$. Next, for any $f : \hom_A(a, a')$, we have for any $b : B$ the morphism $F_{(a,b),(a',b)}(f, 1_b) : F_a(b) \to F_{a'}(b)$. These are the components of a natural transformation $F_a \to F_{a'}$. Functoriality in a is easy to check, so we have a functor $\hat{F} : A \to C^B$.

Conversely, suppose given $G : A \to C^B$. Then for any $a : A$ and $b : B$ we have the object $G(a)(b) : C$, giving a function $A_0 \times B_0 \to C_0$. And for $f : \hom_A(a, a')$ and $g : \hom_B(b, b')$, we have the morphism

$$G(a')_{b,b'}(g) \circ G_{a,a'}(f)_b = G_{a,a'}(f)_{b'} \circ G(a)_{b,b'}(g)$$

in $\hom_C(G(a)(b), G(a')(b'))$. Functoriality is again easy to check, so we have a functor $\check{G} : A \times B \to C$.

Finally, it is also clear that these operations are inverses. □